Answer Key with Explanations

100 Questions - 200 Marks - Negative marking -0.5. Work each problem before checking. Where a question has an ambiguous published key, the explanation states the intended logic.

SECTION A - General Intelligence & Reasoning

Q1. Answer: (b)

Pattern: x2 + (1,2,3,4,5...). 3->7 (x2+1), 7->16 (x2+2), 16->35 (x2+3), 35->74 (x2+4), 74->74x2+5 = 153.

Q2. Answer: (a)

These are n^3 - 1 for n = 2,3,4,5,6: 2^3-1=7, 3^3-1=26, 4^3-1=63, 5^3-1=124, 6^3-1=215. Next: 7^3-1 = 343-1 = 342.

Q3. Answer: (b)

Rule: n*(n+1). 6 -> 6x7 = 42. So 8 -> 8x9 = 72.

Q4. Answer: (a)

Pattern: x2 + 1: 6->13 (x2+1), 13->28 (x2+2), 28->59 (x2+3), 59->122 (x2+4), 122->122x2+5 = 249.

Q5. Answer: (d)

He walks N10, E8 (right from N is E), S4 (right again), then E6 (left). East total = 8+6 = 14, North total = 10-4 = 6; now facing East. Straight-line distance = sqrt(14^2 + 6^2). Both (b) and (d) state this distance, but (d) also gives the facing direction (East), so it is the most complete answer. Answer (d).

Q6. Answer: (a)

Each number n maps to 5^n: 1->5, 2->25, 3->125, 4->625, so 5->5^5 = 3125.

Q7. Answer: (c)

All keys are locks; some locks are doors (the door-locks need not be keys) -> 'some keys are doors' not necessary. No door is window, but locks-windows undetermined. None of I, II, III necessarily follows. Answer (c).

Q8. Answer: (b)

Two-face-painted cubes lie on edges (not corners). For side n: 12(n-2). n=5 -> 12x3 = 36.

Q9. Answer: (a)

First letters: A(+6)G(+6)M(+6)S(+6)Y. Second letters: Z(-6)T(-6)N(-6)H(-6)B. So next pair = YB. Answer (a) YB.

Q10. Answer: (a)

Substitute real operators: - means /, + means x, x means -, / means +. Expression 18 - 3 + 5 x 2 / 4 becomes 18 / 3 x 5 - 2 + 4. BODMAS: 18/3 = 6; 6 x 5 = 30; then 30 - 2 + 4 = 32. Answer (a) 32.

Q11. Answer: (b)

Grandfather's only son = the man's father. The lady is the daughter of the man's father => she is the man's sister. Answer (b) Sister.

Q12. Answer: (d)

Rule: (a + b) x c. Row1: (2+3)x4 = 20. Row2: (3+4)x5 = 35. Row3: (4+5)x6 = 54. Answer (d) 54.

Q13. Answer: (c)

Every mother is a woman (Mothers subset Women). Doctors can be men or women, so Doctors intersects both Women and Mothers. Answer (c).

Q14. Answer: (c)

Faulty clock loses 5 min/hr so it shows 55 min for every 60 real min -> it is SLOW. When faulty shows 6 hrs (360 faulty min), real elapsed = 360 x (60/55) = 392.7 min = 6 h 32.7 min -> ~6:33 PM. Answer (c).

Q15. Answer: (a)

Terms = n^2(n+1): n=1 ->1x2=2, n=2 ->4x3=12, n=3 ->9x4=36, n=4 ->16x5=80, n=5 ->25x6=150, n=6 ->36x7=252. Answer (a).

Q16. Answer: (a)

Row of 5 facing North (left->right). R is extreme left (pos1), S extreme right (pos5), P middle (pos3), T immediately right of P (pos4). Only Q is left for pos2. So second from left = Q. Answer (a).

Q17. Answer: (d)

Letters symmetric about a vertical mirror line: M, O, M. 'MOM' reads the same in a vertical mirror. Answer (d).

Q18. Answer: (a)

A is SW of B, C is SE of B. From A, B is NE; C is due East of A (same southward offset, C further east). Answer (a) East.

Q19. Answer: (b)

343=7^3, 512=8^3, 729=9^3, 1000=10^3 are perfect cubes; 1320 is not. Answer (b).

Q20. Answer: (b)

Q is 11th from right; after interchange P takes Q's place and is 13th from left, so Q was 13th from left. Total = (13 from left) + (11 from right) - 1 = 23. Answer (b).

Q21. Answer: (d)

Rule: x2 + 1. 6 -> 6x2+1 = 13; 13 -> 27; 27 -> 55; 55 -> 111; 111 -> 111x2+1 = 223. The last term should be 223, not 222, so the wrong term is (d) 222.

Q22. Answer: (a)

16x1x9x14x20 = 16x9=144; 144x14=2016; 2016x20=40320. /5 = 8064. Unit digit = 4. Answer (a).

Q23. Answer: (b)

On a standard dice opposite faces sum to 7: opposite of 3 = 7 - 3 = 4. Answer (b) 4.

Q24. Answer: (a)

Pole: L = v*9. Platform: L+240 = v*24. Subtract: 240 = 15v -> v=16 m/s. L = 16x9 = 144 m. Answer (a).

Q25. Answer: (b)

'Son of the wife of my husband's father' = son of the woman's mother-in-law = the woman's husband or his brother. Since he is explicitly not her husband, he is her husband's brother => brother-in-law. Answer (b).

SECTION B - General Awareness

Q26. Answer: (b)

Article 17 abolishes untouchability.

Q27. Answer: (b)

Doctrine of Lapse was applied by Lord Dalhousie (1848-56).

Q28. Answer: (c)

Tropic of Cancer passes through 8 states (Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram). It does NOT pass through Odisha.

Q29. Answer: (b)

Protecting and improving the natural environment is a Fundamental Duty under Article 51A(g).

Q30. Answer: (c)

Nathu La pass (Sikkim) connects India with China (Tibet).

Q31. Answer: (b)

The slogan 'Inquilab Zindabad' was coined by Maulana Hasrat Mohani; popularised by Bhagat Singh.

Q32. Answer: (c)

Rancidity is oxidation of fats/oils by atmospheric oxygen.

Q33. Answer: (a)

The Marble Rocks gorge at Bhedaghat (Jabalpur) is formed by the Narmada river.

Q34. Answer: (b)

Insulin is secreted by the beta cells of the islets of Langerhans in the pancreas.

Q35. Answer: (b)

The 73rd Amendment (1992) gave constitutional status to Panchayati Raj institutions.

Q36. Answer: (c)

Rice is a Kharif (monsoon-sown) crop; wheat, mustard, barley are Rabi.

Q37. Answer: (a)

Battle of Plassey was fought in 1757 (Clive vs Siraj-ud-Daulah).

Q38. Answer: (b)

Hawala is an informal, often illegal, value-transfer system linked to money laundering.

Q39. Answer: (c)

Vitamin K is synthesised by intestinal bacteria.

Q40. Answer: (b)

The Cripps Mission arrived in March 1942.

Q41. Answer: (c)

The Fourth Schedule deals with allocation of seats in the Rajya Sabha.

Q42. Answer: (b)

The SI unit of electrical conductance is the siemens (S).

Q43. Answer: (b)

The Kailasa Temple at Ellora was built by the Rashtrakuta king Krishna I.

Q44. Answer: (b)

Repo rate is the rate at which the RBI lends to commercial banks against government securities.

Q45. Answer: (c)

The liver is the largest gland in the human body.

Q46. Answer: (c)

The Shimla Agreement was signed in 1972 (after the 1971 war).

Q47. Answer: (c)

Black (Regur) soil retains moisture and is ideal for cotton.

Q48. Answer: (b)

Washing soda is sodium carbonate decahydrate, Na2CO3.10H2O.

Q49. Answer: (b)

Akbar abolished the Jizya tax (later re-imposed by Aurangzeb).

Q50. Answer: (b)

The Coriolis force deflects winds and ocean currents (right in N hemisphere, left in S).

SECTION C - Quantitative Aptitude

Q51. Answer: (a)

x+1/x=3 -> (x+1/x)^3 = x^3+1/x^3 + 3(x+1/x). 27 = (x^3+1/x^3) + 9 -> x^3+1/x^3 = 18.

Q52. Answer: (b)

Doubles in 8 yr SI -> interest = principal in 8 yr, so rate*8 = 100% -> 12.5%/yr. To become 5x, interest = 4P -> 400% / 12.5 = 32 years.

Q53. Answer: (b)

SI for 2 yr = 400 -> 200/yr. CI - SI for 2 yr = 410-400 = 10 = interest on first-year interest = 200 * r. r = 10/200 = 5%.

Q54. Answer: (a)

1/(A+B)=1/12, 1/(B+C)=1/15, 1/(C+A)=1/20. Sum = 2(A+B+C) rate = 1/12+1/15+1/20 = (5+4+3)/60 = 12/60 = 1/5. So (A+B+C)=1/10. All three: 10 days.

Q55. Answer: (a)

h/x... at 30deg: h = d*tan30 from far; standard result h = (40 * tan30 * tan60)/(tan60 - tan30). = 40*(1/sqrt3)*sqrt3 /(sqrt3 - 1/sqrt3) = 40 / (2/sqrt3) = 20*sqrt3 m.

Q56. Answer: (a)

sec+tan=2 and sec-tan=1/2 (since sec^2-tan^2=1). Add: 2sec=2.5 -> sec=1.25 -> cos=0.8 -> sin=0.6 = 3/5. Answer (a) 3/5.

Q57. Answer: (a)

Sum of 11 = 550. First 6 sum = 294, last 6 sum = 312. 6th counted twice: 294+312 - 550 = 56.

Q58. Answer: (a)

MP = 1.4 CP; SP = 0.9 x 1.4 = 1.26 CP -> 26% profit.

Q59. Answer: (a)

Rates: 1/20 + 1/30 - 1/40 = (6+4-3)/120 = 7/120 per min -> time = 120/7 = 17 1/7 min.

Q60. Answer: (b)

2^3 = 8 = 1 mod 7. 2^100 = 2^99 x 2 = (2^3)^33 x 2 = 1 x 2 = 2 mod 7. Remainder = 2. Answer (b).

Q61. Answer: (a)

Upstream speed = 24/6 = 4 km/h; downstream = 36/6 = 6 km/h. Stream = (6-4)/2 = 1 km/h.

Q62. Answer: (b)

a^2+b^2+c^2 = ab+bc+ca implies (a-b)^2+(b-c)^2+(c-a)^2 = 0 -> a=b=c -> equilateral.

Q63. Answer: (a)

Product = HCF x LCM = 11 x 7700 = 84700. Other = 84700 / 275 = 308.

Q64. Answer: (b)

Milk = 40 L, water = 20 L. For milk:water = 1:2, water must be 80 L (twice milk). Add 80-20 = 60 L.

Q65. Answer: (a)

Tn = Sn - S(n-1). S10 = 300+50 = 350; S9 = 243+45 = 288. T10 = 350-288 = 62.

Q66. Answer: (a)

Inradius of equilateral triangle = side/(2*sqrt3) = 12/(2*sqrt3) = 6/sqrt3 = 2*sqrt3 cm.

Q67. Answer: (b)

Required reduction % = (25/(100+25)) x 100 = 25/125 x100 = 20%.

Q68. Answer: (a)

Form a^3+b^3 / (a^2-ab+b^2) = a+b = 0.87+0.13 = 1.

Q69. Answer: (b)

Relative speed = 72-9 = 63 km/h = 63 x 5/18 = 17.5 m/s. Length = 17.5 x 20 = 350 m. Answer (b) 350.

Q70. Answer: (a)

Volume factor = 1.2^2 x 0.8 = 1.44 x 0.8 = 1.152 -> +15.2%.

Q71. Answer: (b)

log(2^64) = 64 x 0.3010 = 19.264. Digits = floor(19.264)+1 = 20.

Q72. Answer: (a)

Capital-months: A = X x12 = 12X; B = 2X x6 = 12X. Ratio 1:1.

Q73. Answer: (b)

20 CP = 16 SP -> SP/CP = 20/16 = 1.25 -> 25% gain.

Q74. Answer: (a)

Sides 13,14,15: s=21, area = sqrt(21x8x7x6) = sqrt(7056) = 84 cm^2.

Q75. Answer: (a)

7^x = 343 -> x = 3; 2^y = 64 -> y = 6. x^2 + y^2 + 2xy = (x + y)^2 = (3 + 6)^2 = 81. Answer (a) 81.

SECTION D - English Comprehension

Q76. Answer: (b)

EPHEMERAL = short-lived; opposite = PERMANENT.

Q77. Answer: (b)

Correct spelling: CONSENSUS.

Q78. Answer: (b)

OBSEQUIOUS = excessively obedient/fawning = SERVILE.

Q79. Answer: (b)

'Neither' is singular -> 'has submitted', not 'have submitted'. Error in (b).

Q80. Answer: (b)

'Let the cat out of the bag' = to reveal a secret.

Q81. Answer: (b)

Correct collocation: 'unanimous IN its decision'.

Q82. Answer: (a)

Present-perfect passive: 'My bicycle has been stolen.'

Q83. Answer: (b)

PARSIMONIOUS = stingy; antonym = GENEROUS.

Q84. Answer: (a)

Correct punctuation: "Where," she asked, "are you going?"

Q85. Answer: (b)

A person who speaks many languages = POLYGLOT.

Q86. Answer: (a)

COGENT = convincing/forceful; fits 'no one could refute it.'

Q87. Answer: (b)

'A storm in a teacup' = much fuss over a trivial matter.

Q88. Answer: (b)

'Each' is singular -> 'was given', not 'were given'. Error in (b).

Q89. Answer: (b)

MITIGATE = to lessen = ALLEVIATE.

Q90. Answer: (b)

Reported speech: present continuous -> past continuous. 'She said that she was reading a book.'

Q91. Answer: (a)

Correct: 'averse TO taking risks.'

Q92. Answer: (b)

Correct spelling: MAINTENANCE.

Q93. Answer: (b)

That which cannot be corrected = INCORRIGIBLE.

Q94. Answer: (b)

GREGARIOUS = sociable; opposite = RECLUSIVE.

Q95. Answer: (b)

Passage: internet is 'now a prerequisite for full participation in modern... life.'

Q96. Answer: (b)

Author: policymakers 'frame the issue purely in terms of infrastructure' while affordability/literacy are equally decisive.

Q97. Answer: (b)

Calling it 'a single technological fix' (dismissively) implies technology alone is insufficient.

Q98. Answer: (b)

'Persistent' = continuing/ENDURING.

Q99. Answer: (b)

Tone is analytical and concerned.

Q100. Answer: (b)

Best title: 'Bridging the Digital Divide: Beyond Infrastructure.'